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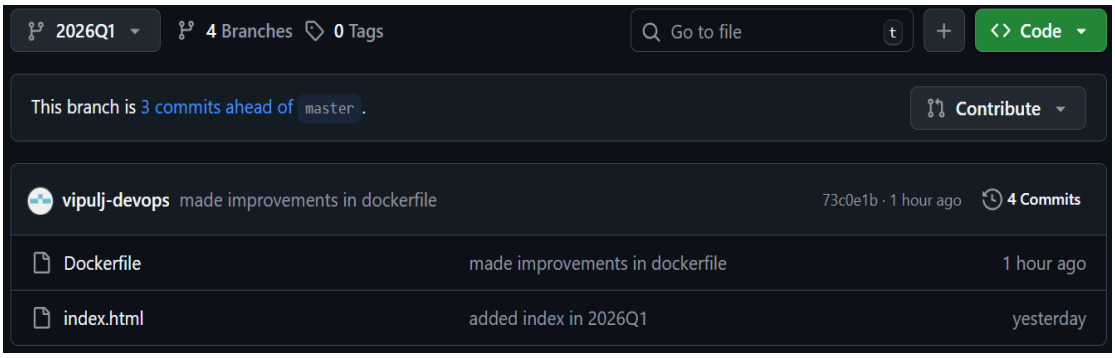
Assignment 1

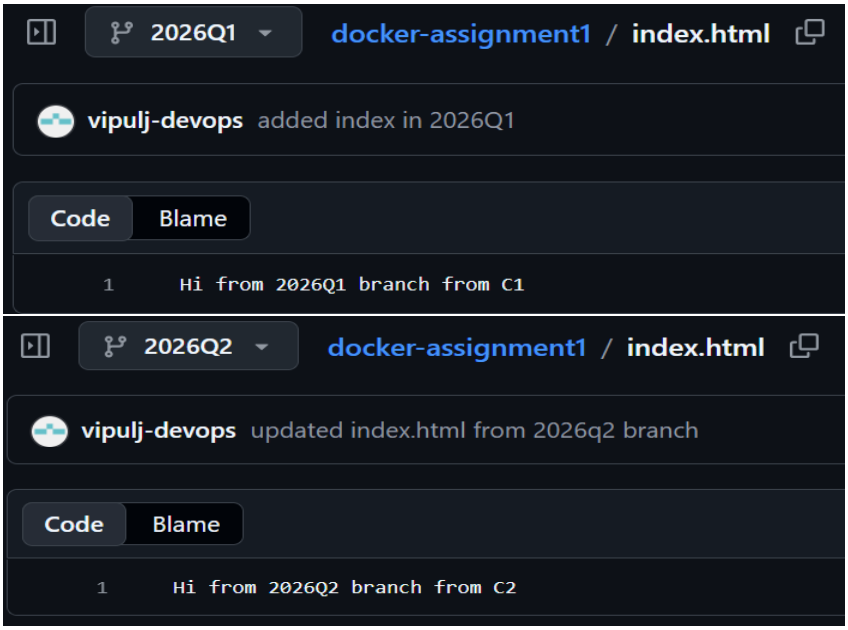
Task: using a Jenkins Declarative Pipeline to build and deploy Docker containers from a Git repository with three branches: 2026Q1, 2026Q2, and 2026Q3, each containing an index.html file. Three containers (c1, c2, and c3) are to be created, where each container hosts the respective branch's index.html file on an Apache HTTP server (httpd). Container c1 should run on port 80, c2 on port 90, and c3 on port 8080 using port binding, with the entire process automated through Jenkins.

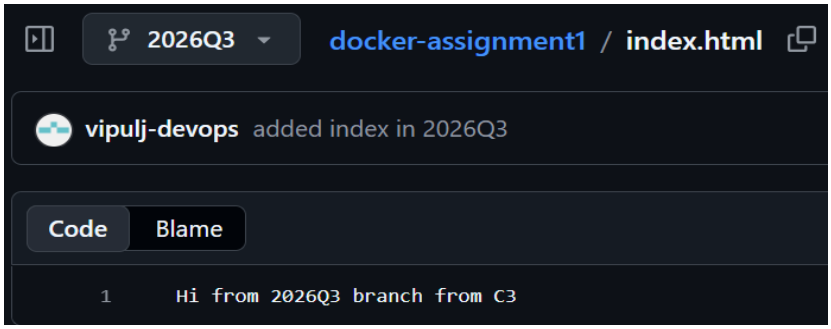
Steps:

1. Created a repository and added index.html and Dockerfile in it

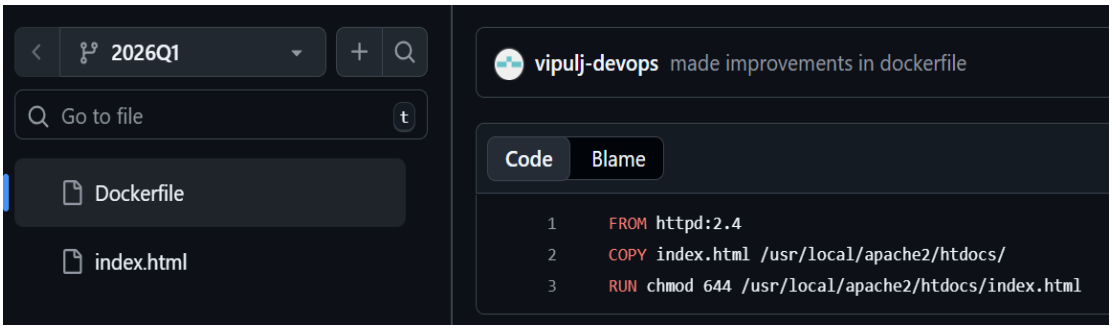


2. Added index.html file in all 3 repositories

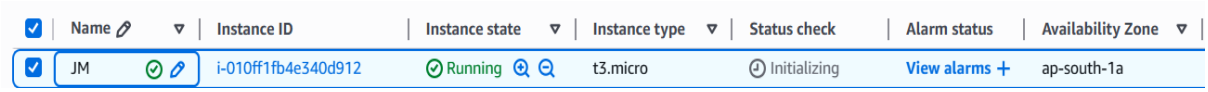




3. Added same Dockerfile in all 3 repositories



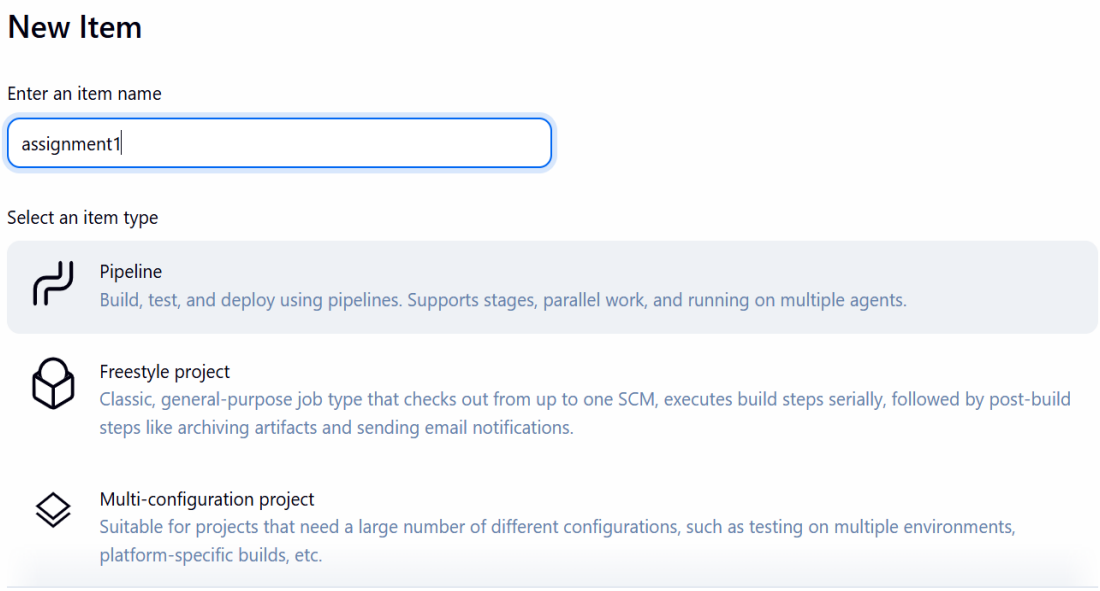
4. Launched Jenkins master container



Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone
JM	i-010ff1fb4e340d912	Running	t3.micro	Initializing	View alarms +	ap-south-1a

5. Installed java-17, docker, tomcat and Jenkins in Jenkins master

6. Created declarative pipeline job in Jenkins master



7. Added Groovy script in the job

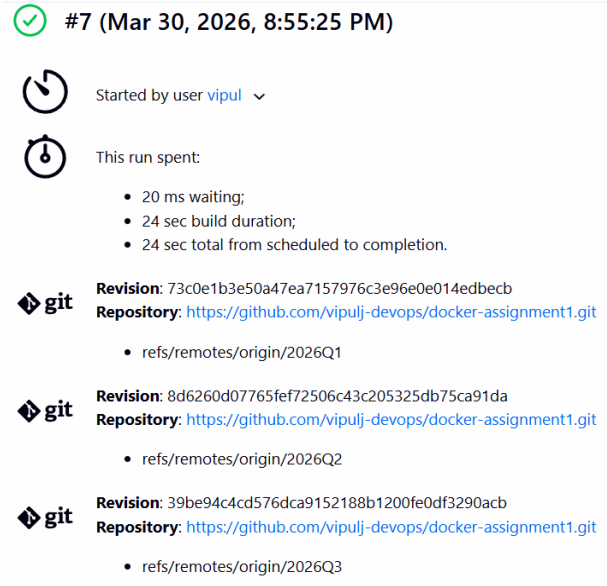
Script:

```
pipeline {
  agent any

  stages {

    stage('Clean Workspace') {
      steps {
        deleteDir()
        sh '''
        docker rm -f c1 c2 c3 || true
        docker system prune -f || true
        '''
      }
    }

    stage('Build & Run Q1') {
      steps {
        git branch: '2026Q1',
        url: 'https://github.com/vipulj-devops/docker-assignment1.git'

        sh '''
        docker build -t image_q1 .
        docker run -d -p 80:80 --name c1 image_q1
        '''
      }
    }

    stage('Build & Run Q2') {
      steps {
        deleteDir()

        git branch: '2026Q2',
        url: 'https://github.com/vipulj-devops/docker-assignment1.git'

        sh '''
        docker build -t image_q2 .
        docker run -d -p 90:80 --name c2 image_q2
        '''
```

```
            }
        }

        stage('Build & Run Q3') {
            steps {
                deleteDir()

                git branch: '2026Q3',
                url: 'https://github.com/vipulj-devops/docker-assignment1.git'

                sh '''
                docker build -t image_q3 .
                docker run -d -p 8081:80 --name c3 image_q3
                '''
            }
        }
    }
}
```

Description: This Jenkins Declarative Pipeline automates the process of building and deploying Docker containers from different branches of a Git repository. It starts by cleaning the workspace and removing any existing containers and unused Docker resources to avoid conflicts. In each stage, the pipeline checks out a specific branch (2026Q1, 2026Q2, and 2026Q3), builds a Docker image using the corresponding index.html file, and runs a container from that image. Each container is deployed on an Apache HTTP server and exposed on different ports using port binding—port 80 for c1, port 90 for c2, and port 8081 (since 8080 was occupied by Jenkins) for c3. The use of deleteDir() ensures a clean workspace before processing each branch, preventing file conflicts between stages.

8. The job ran successfully



9. The images and containers got created

Images:

```
[root@ip-172-31-43-203 ~]# docker images
REPOSITORY    TAG       IMAGE ID       CREATED          SIZE
image_q2      latest    8a1b025844a6   11 hours ago     117MB
image_q3      latest    6af9b6047f50   11 hours ago     117MB
image_q1      latest    a81ceb73ed7b   11 hours ago     117MB
```

Containers:

```
[root@ip-172-31-43-203 ~]# docker ps
CONTAINER ID   IMAGE     COMMAND              CREATED          STATUS          PORTS                                      NAMES
79c71f46be0d   image_q3  "httpd-foreground"   23 minutes ago   Up 23 minutes   0.0.0.0:8081->80/tcp, :::8081->80/tcp      c3
29f412c9864d   image_q2  "httpd-foreground"   23 minutes ago   Up 23 minutes   0.0.0.0:90->80/tcp, :::90->80/tcp          c2
52b3ec021fd1   image_q1  "httpd-foreground"   23 minutes ago   Up 23 minutes   0.0.0.0:80->80/tcp, :::80->80/tcp          c1
```

10. Index.html got deployed successfully on respective ports